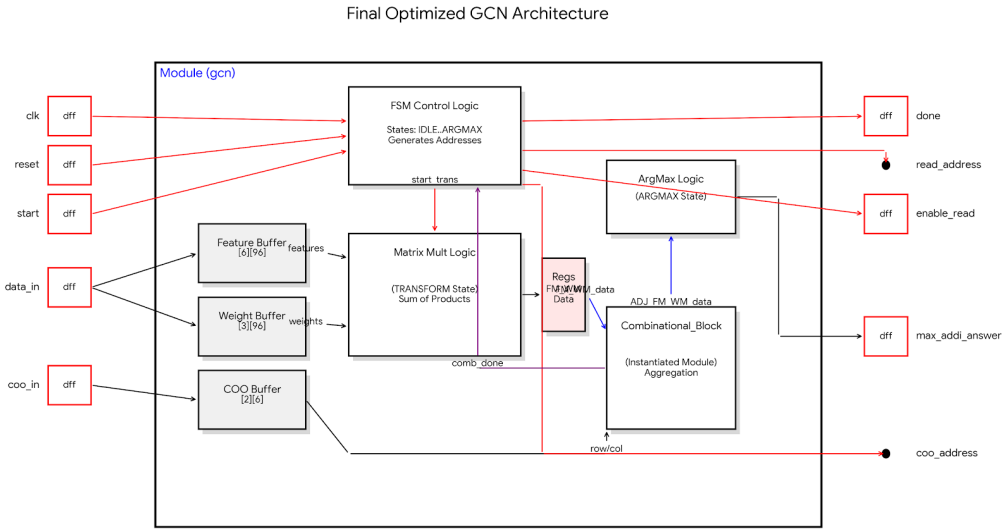
ASIC IMPLEMENTATION

Graph convolutional network accelerator

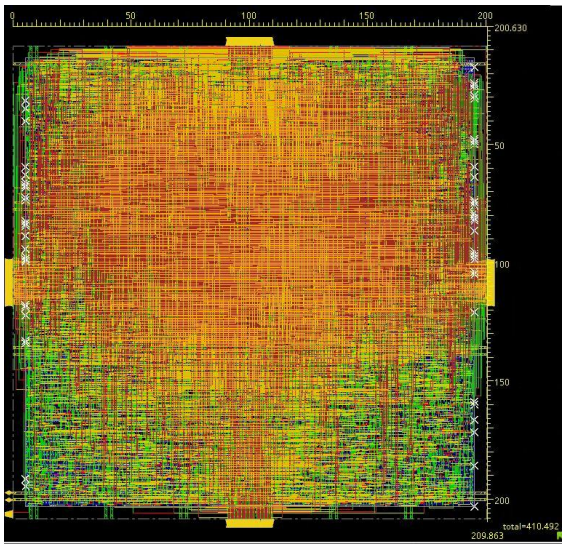
Ruthvik Reddy Anthireddigari / Arizona State University / Spring 2026

I implemented a node classifier with an RTL datapath and a finite-state controller. The same testbench reproduced the reference classifications before synthesis, after synthesis and after routing.

Datapath and control



From synthesized logic to routed layout



The controller sequences feature loading, matrix transformation, neighbor aggregation and selection of the predicted class.

Synopsys Design Compiler synthesized the RTL. Cadence Innovus placed and routed the design. Post-route checks included setup and hold timing.